

Assignment - 7.3

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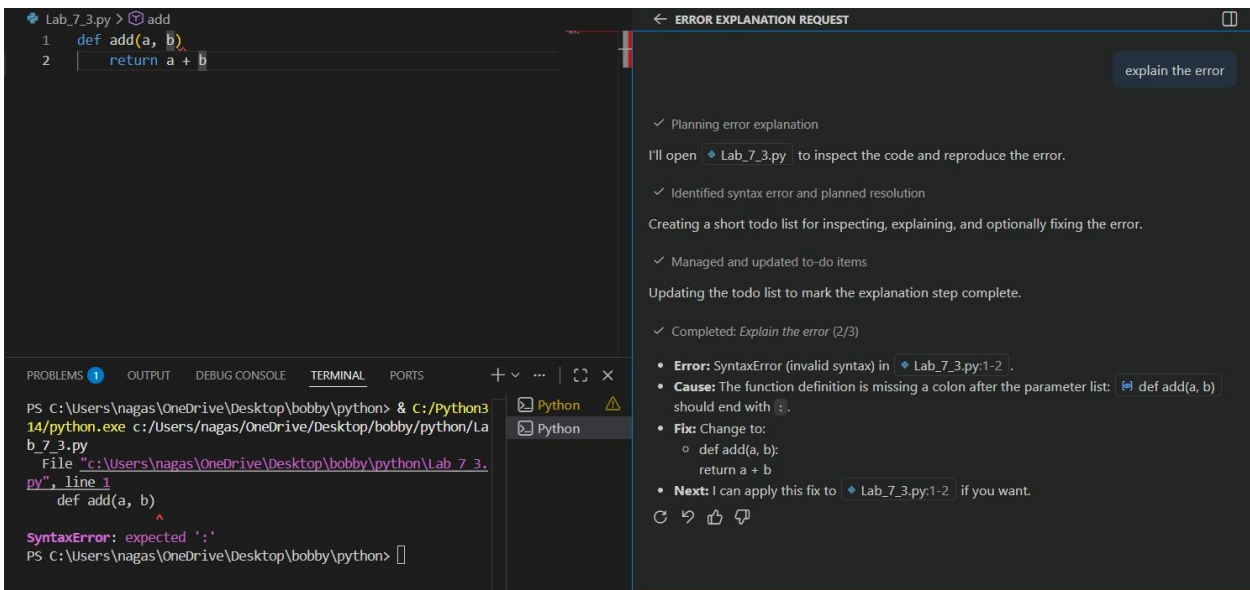
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Task 1: Fixing Syntax Errors

Buggy Code (Before Debugging):

```
def add(a, b)
    return a + b print(add(3,
5))
```

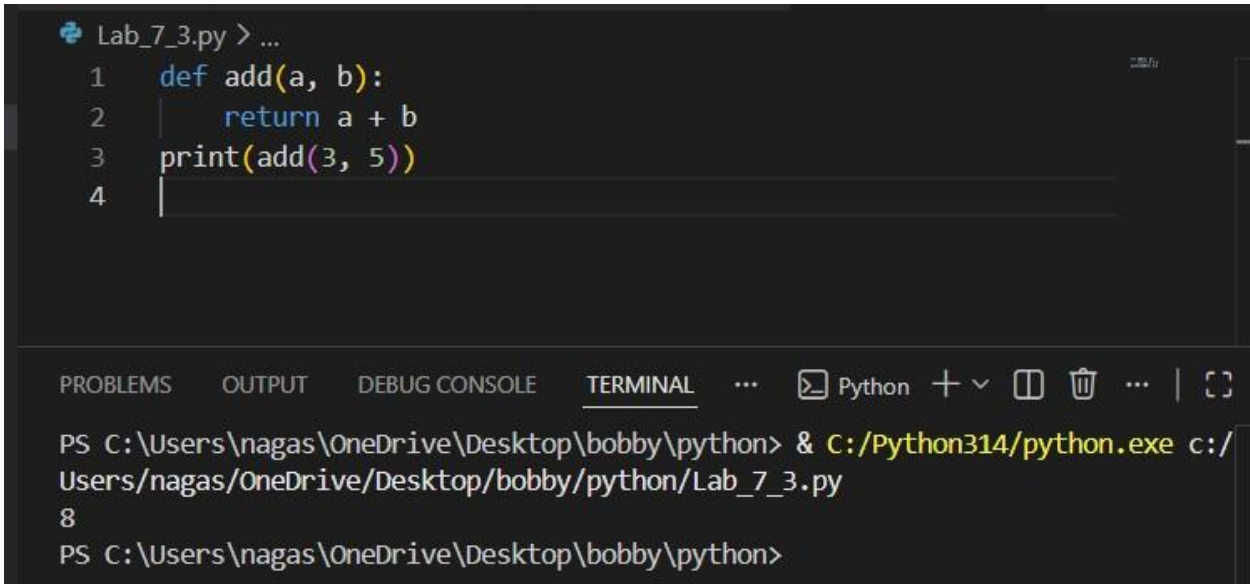


AI-Detected Issue

- Missing colon (:) at the end of the function definition.
- This causes a **SyntaxError**.

Corrected Code (After Debugging):

```
def add(a, b):  
    return a + b  
print(add(3, 5))
```



The screenshot shows a code editor window with a file named 'Lab_7_3.py'. The code in the editor is:

```
1 def add(a, b):  
2     return a + b  
3 print(add(3, 5))  
4
```

Below the editor is a terminal window. The terminal shows the command to run the script using Python 3.14:

```
PS C:\Users\nagas\OneDrive\Desktop\bobby\python> & C:/Python314/python.exe c:/Users/nagas/OneDrive/Desktop/bobby/python/Lab_7_3.py  
8  
PS C:\Users\nagas\OneDrive\Desktop\bobby\python>
```

AI Explanation

In Python, every function definition must end with a colon.

The missing colon caused the syntax error. Adding it fixes the issue.

Task 2: Debugging Logic Errors in Loops

Buggy Code (Before Debugging):

...

```

    ↕ 1 frames
/usr/local/lib/python3.12/dist-packages/ipykernel/iostream.py in _is_master_process(self)
    436
    437     def _is_master_process(self):
--> 438         return os.getpid() == self._master_pid
    439
    440     def set_parent(self, parent):

```

KeyboardInterrupt:

```

1 i = 1
2 while i <= 5:
3     print(i)
4
5 + i += 1

```

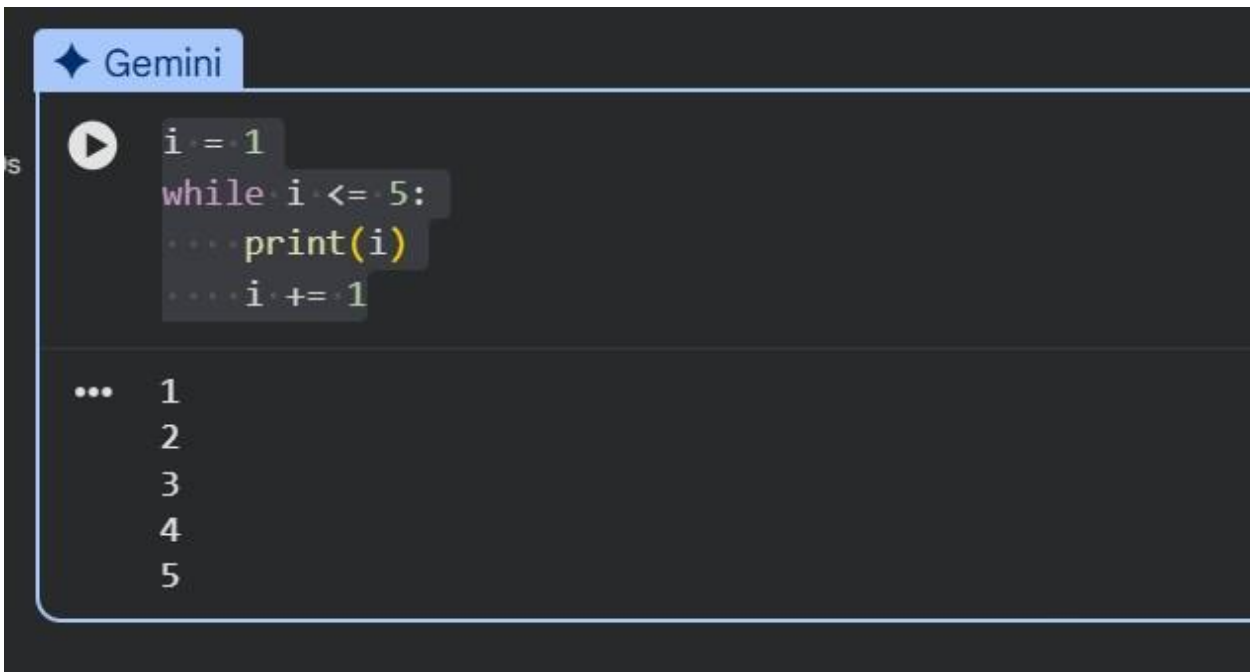
Gemini

It looks like the `while` loop was stuck in an infinite loop because the variable `i` was never incremented. I'll modify the cell to include `i += 1` so the loop can terminate correctly.

AI-Detected Issue

- The loop variable `i` is never incremented.
- This causes an **infinite loop**.

Corrected Code (After Debugging)



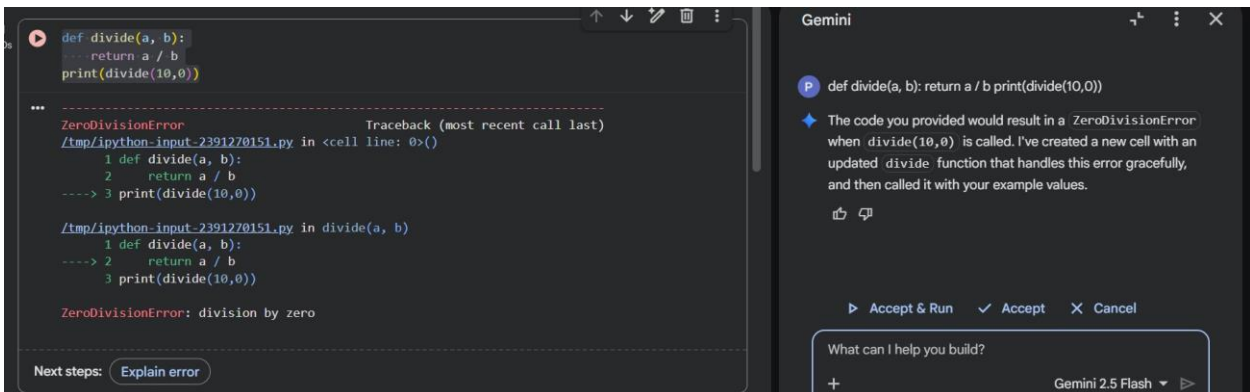
The screenshot shows a code editor with a dark theme. At the top, there's a blue tab labeled "Gemini". Below it, a code cell contains the following Python code:

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

Below the code cell, the output is displayed as a list of numbers from 1 to 5, each on a new line.

Task 3: Handling Runtime Errors (Division by Zero) Buggy

Code (Before Debugging):



The screenshot shows a code editor with a dark theme. On the left, a code cell contains the following Python code:

```
def divide(a, b):
    return a / b
print(divide(10,0))
```

Below the code cell, the output is displayed as a traceback for a `ZeroDivisionError`:

```
ZeroDivisionError: division by zero
Traceback (most recent call last):
  /tmp/ipython-input-2391270151.py in <cell line: 0>()
    1 def divide(a, b):
    2     return a / b
----> 3 print(divide(10,0))

/tmp/ipython-input-2391270151.py in divide(a, b)
    1 def divide(a, b):
----> 2     return a / b
    3 print(divide(10,0))

ZeroDivisionError: division by zero
```

Below the error message, there's a "Next steps:" section with a button labeled "Explain error".

On the right, there's a Gemini chat window. It shows a prompt "P" with the code `def divide(a, b): return a / b print(divide(10,0))` and a response "♦" explaining the error and providing a corrected version of the code.

AI-Detected Issue

- If `b` is zero, the program crashes with **ZeroDivisionError**.

Corrected Code (After Debugging):

```
def divide(a, b):
    try:
        return a / b
    except ZeroDivisionError:
        return "Error: Cannot divide by zero"

print(divide(10,0))
```

... Error: Cannot divide by zero

AI Explanation

The AI used a `try-except` block to handle runtime errors safely. This prevents the program from crashing.

Task 4: Debugging Class Definition Errors

Buggy Code (Before Debugging):

```
class Student:
    def __init__(name, age):
        name = name
        age = age
    def __init__(self, name, age):
        self.name = name
        self.age = age
s = Student("Rahul", 20)
print(s.name, s.age)
```

TypeError: Student.__init__() takes 2 positional arguments but 3 were given

Traceback (most recent call last):
 File "/tmp/ipython-input-1522557614.py", line 0:()
 3 name = name
 4 age = age
----> 5 s = Student("Rahul", 20)
 6 print(s.name, s.age)

TypeError: Student.__init__() takes 2 positional arguments but 3 were given

Next steps: [Explain error](#)

Gemini

TypeError: Student.__init__() takes 2 positional arg

The error `TypeError: Student.__init__() takes 2 positional arguments but 3 were given` means that the `__init__` method was called with more arguments than it was defined to accept. This is because all instance methods in Python, including `__init__`, automatically receive the instance itself as the first argument, conventionally named `self`. Your `__init__` method was missing this `self` parameter. Also, the `name` and `age` parameters were not correctly assigned to the `Student` object's attributes. I'll correct the `__init__` method to properly define `self` and assign the values.

What can I help you build?

Gemini 2.5 Flash

Gemini can make mistakes so double-check it and use code with caution. [Learn more](#)

AI-Detected Issue

- Missing `self` parameter in the constructor.
- Instance variables are not assigned correctly.

Corrected Code (After Debugging):

```
◆ Gemini

▶ class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age
s = Student("Rahul", 20)
print(s.name, s.age)

... Rahul 20
```

Task 5: Resolving Index Errors in Lists Buggy

Code (Before Debugging):

```
▶ numbers = [10, 20, 30]
print(numbers[5])

...
-----
IndexError                                Traceback (most recent call last)
/tmp/ipython-input-2134759640.py in <cell line: 0>()
      1 numbers = [10, 20, 30]
----> 2 print(numbers[5])

IndexError: list index out of range

Next steps: Explain error
```

Gemini

P Please explain this error:

ⓘ **IndexError: list index out of range**

◆ The error `IndexError: list index out of range` occurs because you are trying to access the element at index 5 in the `numbers` list, but the list only has three elements (at indices 0, 1, and 2). To fix this, you should use a valid index. For example, if you want to print the last element, you would use `numbers[2]`.

AI-Detected Issue

- Index 5 does not exist in the list.
- Causes an **IndexError**.

Corrected Code (After Debugging):

```
numbers = [10, 20, 30]
print(numbers[0])

... 10
```

Gemini

P Print a valid index from the numbers list

◆ Certainly! I'll modify the existing cell to print an element using a valid index. For example, here's how to print the first element:

The code executed successfully, and as requested, it printed the element at index `0` from the `numbers` list, which is `10`.

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